

Virtual Reality Therapy for PTSD Patients: A Database System for Monitoring Sessions and Tracking Therapeutic Progress

1. Executive Summary

This project focuses on the development of a relational database system that supports the tracking and analysis of therapy sessions conducted in virtual reality (VR) environments for patients suffering from Post-Traumatic Stress Disorder (PTSD). The aim is to create a system that efficiently records patient data, VR therapy session details, clinician involvement, trauma scenarios, and therapy outcomes over time.

Key Features:

- Structured storage of patient and clinician information
- VR scenario management for trauma exposure
- Session tracking with outcome evaluations
- Progress monitoring through pre- and post-session scores

Expected Outcomes:

- Improved monitoring of therapeutic effectiveness
- Reliable historical data for clinicians and researchers
- Better organization and retrieval of therapy session information

2. Actors and Users

Administrator

- Manages user accounts and permissions
- Oversees database integrity and backups

Clinician

- Records patient sessions and progress
- Assigns VR scenarios and evaluates outcomes

Patient (optional)

- May view personal progress reports or attend VR therapy sessions

Each actor interacts with the system through defined permissions and access levels to ensure data integrity and confidentiality.

3. System Overview and Requirements

System Purpose:

To support mental health professionals by offering a structured database system that facilitates the recording, tracking, and analysis of PTSD therapy administered via virtual reality.

Functional Requirements:

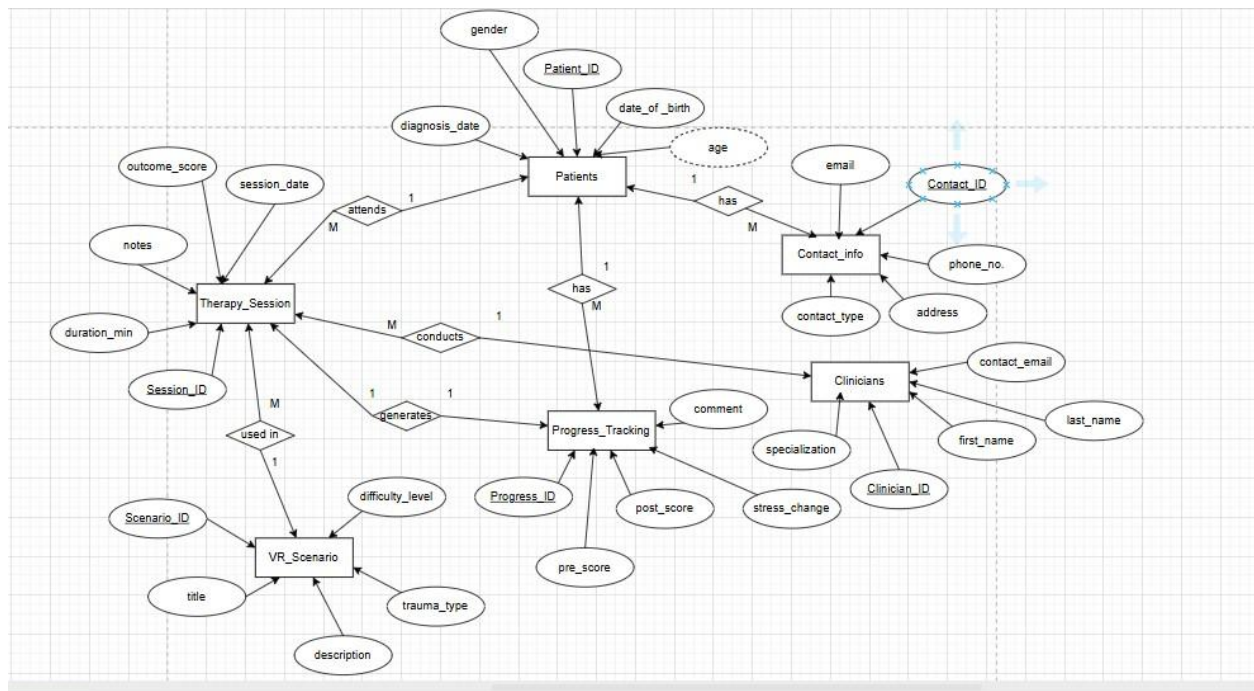
- Add/view/update/delete patient records
- Schedule and record therapy sessions
- Assign VR trauma scenarios
- Track and evaluate therapy outcomes
- Store emergency contact and clinician details

Non-Functional Requirements:

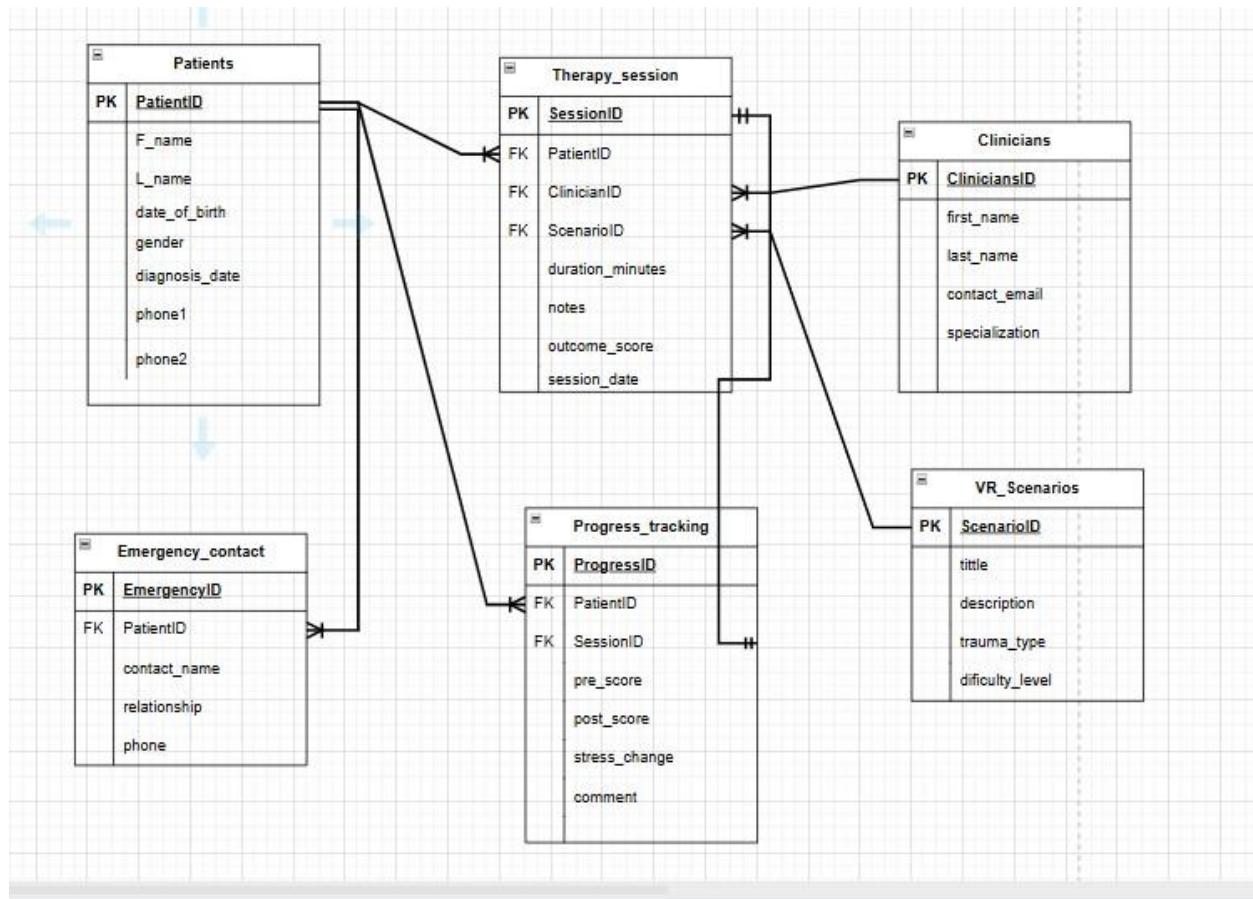
- Secure access with user roles
 - Scalable relational structure
 - User-friendly interface for clinicians
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4. Conceptual Database Design

ER DIAGRAM



RELATIONAL MODEL



Entities and Attributes:

- **Patients:** patient_id, first_name, last_name, date_of_birth, gender, diagnosis_date
- **ContactInfo:** , patient_id, contact_type, contact_detail
- **EmergencyContacts:** EmergencyID, Patient_id, name, relationship, phone_number
- **Clinicians:** clinician_id, first_name, last_name, specialization, contact_email
- **VRScenario:** scenario_id, title, description, trauma_type, difficulty_level
- **TherapySessions:** session_id, patient_id, session_date, duration, notes, outcome_score, clinician_id, scenario_id

- **ProgressTracking:** progress_id, session_id, pre_score, post_score, stress_change, comments

5. Normalization Process

UNNORMALIZED TABLE

A flat table combining all data from patients, clinicians, sessions, scenarios, progress, and contact details, containing redundancy and multivalued attributes, with a total of 22 columns

Patient ID	Full_Name	DOB	Age	Gender	Diagnosis_Date	Contact_Info
P001	Amina Suleman	15/06/1990	33	Female	01/05/2024	0595689254, 0245689523
P002	Kwame Boateng	22/09/1985	38	male	07/06/2024	026 5879 856
P001	Amina Suleman	15/06/1990	33	Female	08/06/2024	0595689254, 0245689523
P003	Sarah Owusu	10/03/1992	32	Female	01/01/2025	055 3562 584

Emergency_Contact	SessionID	Session_Date	Duration_Minutes	Notes	Outcome_Score	ClinicianID
Rahim- brother- 0248765235	S001	19/01/2025	45	Mild response to therapy	5	C001
Ama-wife- 0557845623	S003	23/03/2025	40	Resistance to scenario	4	C002
Rahim- brother- 0248765235	S002	20/05/2025	50	Strong emotional relief	7	C001
Kofi-father- 0265894521	S004	18/06/2025	60	Neutral reaction	6	C002

Full_Name	Specialization	Contact_email	ScenarioID	Title	Description
2					
Dr. Mensah Aboagye	Trauma specialist	mensah@clinic.org	VRS01	war flashback	simulates combat zone
Dr. Linda Owusu	Clinical psychologist	linda@therapy.org	VRS01	war flashback	simulate combat zone

Dr. Mensah Aboagye	Trauma specialist	mensah@clinic.org	VRS02	fire rescue	simulate combat zone
Dr. Linda Owusu	Clinical psychologist	linda@therapy.org	VRS03	accident recovery	simulates car crash trauma

Trauma_type	Difficulty_level	ProgressID	Pre_score	Post_score	Stress_change	Comments
Military	High	PT001	8	5	-3	Patients showed mild improvement
Military	High	PT003	9	8	-1	Very minimal improvement
Fire	Medium	PT002	7	3	-4	Patient highly relieved
Accident	Medium	PT004	6	4	-2	Slight drop in stress level

FIRST NORMAL FORM (1NF):

- Removed multivalued fields: contact_info split into multiple columns.
NB: Patients are to provide a maximum of two phone numbers.
- Emergency contact split into structured fields (name, relationship, phone).
- Separate full_name into First_name and Last_name
- Resulting tables: Combined table, and Emergency_Contacts

Resulting tables after 1NF

Combined Table

PatientID	F_Name	L_Name	Age	Gender	Phone1	Phone2
P001	Amina	Suleman	33	Female	059 5689 254	024 5689 523

P002	Kwame	Boateng	38	Male	026 5879 856	Null
P001	Amina	Owusu	33	Female	059 5689 254	024 5689 523
P003	Sarah	Owusu	32	Female	055 3562 584	Null

SessionID	Session_date	Duration_minutes	Notes	Outcome_score	ClinicianID	F_name
S001	19/01/2025	45	Mild response to therapy	5	C001	Dr. Mensah
S003	23/03/2025	40	Resistance to scenario	4	C002	Dr. Linda
S002	20/05/2025	50	Strong emotional relieve	7	C001	Dr. Mensah
S004	18/06/2025	60	Neural reaction	6	C002	Dr. Linda

L_name	Specialization	Contact_email	ScenarioID	Title	Description	Trauma_Type
Aboagye	Trauma specialist	mensah@clinic.org	VRS01	War flashback	Stimulates combat zone	Military
Owusu	Clinician psychologist	linda@therapy.org	VRS01	War flashback	Stimulates combat zone	Military
Aboagye	Trauma specialist	mensah@clinic.org	VRS02	Fire rescue	Stimulates being trapped	Fire
Owusu	Clinician psychologist	linda@therapy.org	VRS03	Accident recovery	Stimulates car crash trauma	Accident

Difficulty_level	ProgressID	Pre_score	Post_score	Stress_Change	Comment
High	PT001	8	5	-3	Patient showed mild improvement
High	PT003	9	8	-1	Very minimal improvement
Medium	PT002	7	3	-4	Patient highly relieved

medium	PT004	6	4	-2	Slight drop in stress level
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Emergency Contact Table

EmergencyID (PK)	PatientID (FK)	contact_name	relationship	Phone
EC001	P001	Rahim	brother	024 8765235
EC002	P002	Ama	wife	055 7845 623
EC003	P003	Kofi	father	026 5894 521

SECOND NORMAL FORM (2NF):

- Removed partial dependencies:
 - Clinician details moved to Clinicians table
 - Scenario details moved to VRScenario table
- Resulting tables: TherapySessions, Clinicians, VRScenario

Tables after 2NF

SESSION

SessionID	PatientID(FK)	ClinicianID	ScenarioID	Diagnosis_date	Session_date	duration	Outcome_score
S001	P001	C001	VRS01	01/05/2024	19/01/2025	45	5
S003	P002	C002	VRS01	07/06/2024	23/03/2025	40	4
S002	P001	C001	VRS02	08/06/2024	20/05/2025	50	7
S004	P003	C002	VRS03	01/01/2025	18/06/2025	60	6

PATIENT

PatientID	F_name	L_name	DOB	Age	gender	Phone1	Phone2
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P001	Amina	Suleman	15/06/1990	33	Female	05956254	0245689523
P002	Kwame	Boateng	22/09/1985	38	Male	0265879856	Null
P003	Sarah	Owusu	10/03/1992	32	female	0553562584	Null

PROGRESS

ProgressID	Session(FK)	Patient(FK)	Pre_Score	Post_Score	Comments
PT001	S001	P001	8	5	Patient showed mild improvement
PT003	S003	P002	9	8	Very minimal improvement
PT002	S002	P001	7	3	Patient highly relieved
PT004	S004	P003	6	4	Slight drop in stress level

CLINICIANS

ClinicianID	F_Name	L_Name	Specialization	Contact Email
C001	Dr. Mensah	Aboagye	Trauma specialist	mensah@clinic.org
C002	Dr. Linda	Owusu	Clinical psychologist	linda@therapy.org
C001	Dr. Mensah	Aboagye	Trauma specialist	mensah@clinic.org
C002	Dr. Linda	Owusu	Clinical psychologist	linda@therapy.org

VR Scenario

scenarioID	Title	description	trauma_type	difficulty_level
VRS01	War flashback	simulates combat zone	Military	High
VRS02	Fire rescue	simulates being trapped	Military	Medium
VRS03	Accident recovery	simulates car crash trauma	Fire	Medium

Emergency Contact Table

EmergencyID (PK)	PatientID (FK)	contact_name	relationship	Phone
EC001	P001	Rahim	brother	024 8765235
EC002	P002	Ama	wife	055 7845 623
EC003	P003	Kofi	father	026 5894 521

THIRD NORMAL FORM (3NF):

Goals:

1. Remove transitive dependencies
2. Ensure non-key attributes don't depend on other non-key attributes

Process:

1. Identify calculated fields (stress_change), and (Age)
2. Remove derived attributes that can be computed at query time
3. Verify no non-key columns depend on other non-key columns

Tables after 3NF

Progress tracking

Difficulty_level	ProgressID	Pre_score	Post_score	Comment
High	PT001	8	5	Patient showed mild improvement
High	PT003	9	8	Very minimal improvement
Medium	PT002	7	3	Patient highly relieved
Medium	PT004	6	4	Slight drop in stress level

Patient

Patient ID	Full_Name	DOB	Gender	Diagnosis_Date	Contact_Info
P001	Amina Suleman	15/06/1990	Female	01/05/2024	0595689254, 0245689523
P002	Kwame Boateng	22/09/1985	male	07/06/2024	026 5879 856
P001	Amina Suleman	15/06/1990	Female	08/06/2024	0595689254, 0245689523
P003	Sarah Owusu	10/03/1992	Female	01/01/2025	055 3562 584

6. Logical Database Design

Relational Schema:

- Patients(patient_id, first_name, last_name, date_of_birth, gender, diagnosis_date)
- ContactInfo(Patient_id, contact_type, contact_detail)
- EmergencyContacts(emergency_id, patient_id, name, relationship, phone_number)
- Clinicians(clinician_id, first_name, last_name, specialization, contact_email)
- VRScenario(scenario_id, title, description, trauma_type, difficulty_level)
- TherapySessions(session_id, patient_id, session_date, duration, notes, outcome_score, clinician_id, scenario_id)
- ProgressTracking(progress_id, session_id, pre_score, post_score, stress_change, comments)

7. Entity Descriptions

- **Patients:** Stores personal and clinical information about PTSD patients.
- **ContactInfo:** Stores one or more contact methods per patient (e.g., phone, email).
- **EmergencyContacts:** Contains emergency contact details for each patient.
- **Clinicians:** Contains information about mental health professionals.
- **VRScenario:** Holds details about virtual scenarios used for therapy.

- **TherapySessions:** Records each therapy session conducted.
 - **ProgressTracking:** Measures changes in patient condition after sessions.
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